

1. MAXIMUM LIKELIHOOD

Let $X_1, X_2, \dots, X_n$ be independent random variables with density function

$$f(x) = \alpha x^{\alpha-1} \quad \text{for } 0 < x < 1,$$

and $f(x) = 0$ otherwise. The parameter α is positive.

- (a) *Points: 6p.* Derive the maximum likelihood estimator of α .

Solution:

The log-likelihood is

$$\ell(\alpha) = \sum_{i=1}^n \log f(x_i|\alpha),$$

where

$$f(x|\alpha) = \log \alpha + (\alpha - 1) \log x$$

so the log-likelihood is

$$\ell(\alpha) = \sum_{i=1}^n \log f(x_i|\alpha) = \sum_{i=1}^n (\log \alpha + (\alpha - 1) \log x_i) = n \log \alpha + (\alpha - 1) \sum_{i=1}^n \log x_i$$

We find the MLE by setting the derivative of the log-likelihood function to zero and solve for α

$$\ell'(\alpha) = \frac{n}{\alpha} + \sum_{i=1}^n \log x_i = 0$$

which gives the solution

$$\hat{\alpha} = -\frac{n}{\sum_{i=1}^n \log x_i}.$$

Note that it seems that the estimate is negative (because of the minus sign), but it is actually positive since each observation $0 < x_i < 1$ and therefore $\log x_i < 0$ for all x_i . To verify that this indeed a maximum we check that the second derivative of $\ell(\alpha)$ is negative at $\hat{\alpha}$. The second derivative is

$$\ell''(\alpha) = \frac{d}{d\alpha} \ell'(\alpha) = \frac{d}{d\alpha} \left(\frac{n}{\alpha} + \sum_{i=1}^n \log x_i \right) = -\frac{n}{\alpha^2} < 0$$

so this indeed a maximum.

- (b) *Points: 4p.* Determine the usual large sample approximation of the sampling distribution of the maximum likelihood estimator $\hat{\alpha}$.

Solution:

From the Formula sheet we have the normal approximation of the sampling distribution of the MLE

$$\hat{\alpha}(X_1, \dots, X_n) \stackrel{\text{approx}}{\sim} N(\alpha, I_n^{-1}(\alpha))$$

where $I_n(\theta)$ is the Fisher information

$$I_n(\alpha) = \mathbb{E}(-\ell''(\alpha)) = \mathbb{E}\left(\frac{n}{\alpha^2}\right) = \frac{n}{\alpha^2},$$

since $\frac{n}{\alpha^2}$ is a constant (with respect to the data $X_1, \dots, X_n$) so the expectation is just this constant. Hence we have the large sample approximation

$$\hat{\alpha}(X_1, \dots, X_n) \stackrel{\text{approx}}{\sim} N\left(\alpha, \frac{\alpha^2}{n}\right)$$